*International Journal of Hepatology*

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Relevance to HepatoTwin: GGT was selected by both LASSO and Boruta and was among the most influential features in the XGBoost SHAP analysis. We already include LBXSGTSI (GGT), so this provides literature support for retaining it.

RESEARCH ARTICLE OPEN ACCESS

# Machine Learning Models to Predict Metabolic Dysfunction-Associated Steatotic Liver Disease (MASLD) With Simple Anthropometric and Biochemical Variables: A Cross-Sectional Study in US Population

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**Received:** 22 October 2025 | **Revised:** 16 January 2026 | **Accepted:** 14 February 2026

**Academic Editor:** Dirk Uhlmann

**Keywords:** machine learning | metabolic dysfunction-associated steatotic liver disease | National Health and Nutrition Examination Survey

## ABSTRACT

**Background**: Metabolic dysfunction-associated steatotic liver disease (MASLD) is an emerging global health concern. This study was aimed at exploring the feasibility of utilizing machine learning (ML) algorithms to predict MASLD in large general populations based on simple anthropometric and biochemical parameters.

**Methods**: Data from the 2017–2020 cycles of the US National Health and Nutrition Examination Survey (NHANES) were utilized. A total of 6814 participants (53.0% female) with complete transient elastography data were included. MASLD was defined as a controlled attenuation parameter ≥ 280 dB/m, with cardiometabolic risk factor and without excessive alcohol use. Key characteristics and biomarkers associated with MASLD were identified using the least absolute shrinkage and selection operator (LASSO) and the Boruta algorithms. ML methods, including logistic regression (LR), extreme gradient boosting (XGBoost), bootstrap aggregating, random forest, naive Bayes, light gradient boosting machine (LightGBM), decision tree, and support vector machines, were employed to develop the MASLD prediction models.

**Results**: The median age of the 6814 participants was 53 years (interquartile range: 37~65). MASLD was detected among 2611 (38.3%) participants. Key predictors selected via LASSO and Boruta algorithms included body weight, standing height, waist circumference, diagnosis of diabetes, alanine aminotransferase, aspartate aminotransferase, and gamma glutamyl transferase. The areas under the receiver operating characteristic curves of LR, XGBoost, and other ML models were 0.841, 0.837, 0.815, 0.838, 0.814, 0.842, 0.796, and 0.828 in the internal validation cohort. Results indicate that LR, XGBoost, and LightGBM models outperform other models in predicting MASLD.

**Conclusions**: The ML models of LR, XGBoost, and LightGBM are effective and simplified tools for predicting MASLD in the US general population. This study underscores the potential of ML models with simple noninvasive biomarkers in enhancing early detection and personalized management of fatty liver disease.

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## 1 | Background

Metabolic dysfunction-associated steatotic liver disease (MASLD), previously termed metabolic dysfunction-associated fatty liver disease (MAFLD) or nonalcoholic fatty liver disease (NAFLD), has seen a rapid global increase, overtaking viral hepatitis to become the most prevalent chronic liver disorder [1–4]. Studies have shown that MASLD can lead to a variety of more severe conditions, such as advanced fibrosis or cirrhosis [3], metabolic and cardiovascular diseases [5], hepatocellular carcinoma (HCC), or even mortality [5–7]. Early detection of MASLD is therefore critical, facilitating the implementation of appropriate medical and lifestyle interventions.

The liver biopsy was considered the definitive standard for diagnosing MASLD; however, its invasive nature, high cost, and potential procedure-related complications limit its routine use [8]. Consequently, noninvasive markers, including blood tests and imaging techniques, have been increasingly applied to assess fatty liver disease [9, 10]. For example, higher alanine aminotransferase (ALT)/aspartate aminotransferase (AST) ratio may be independently associated with a significantly higher risk of NAFLD among American cohorts [11]. There are studies showing the association of a higher controlled attenuation parameter (CAP), which indicates the presence of hepatic steatosis, with increased age, body mass index (BMI), waist-to-hip ratio, diabetes, hypertension, ALT, and serum uric acid-to-high-density lipoprotein cholesterol ratio [12, 13]. Triglyceride glucose (TyG), TyG-BMI, and TyG-waist circumference indices have been developed and validated to predict both the prevalence and mortality of MASLD [14–16]. Despite these advances, a simple and effective noninvasive screening tool for MASLD remains limited.

Recent advances in machine learning (ML) have significantly expanded the possibilities for predicting the risk of MASLD using demographic, clinical, and biochemical factors [4, 17]. ML algorithms, such as logistic regression (LR), random forest (RF), and extreme gradient boosting (XGBoost), have demonstrated potential in modeling complex, nonlinear relationships between predictors and outcomes [18–20]. These models can incorporate a broad spectrum of variables and achieve higher prediction accuracy compared to traditional statistical methods. However, many existing predictive models were established for specific populations [21] or fibrosis among MASLD patients, which restricts their generalizability across broader contexts [22, 23]. There are few MASLD prediction ML models tailored to general populations [20, 24, 25]. Also, some ML models for the general population included too many predictors or without liver-related biomarkers, affecting the wide use or accuracy [20, 26]. Even though the liver ultrasound CAP measures could be used for the noninvasive diagnosis of MASLD, the availability and usage of liver ultrasound CAP are restrained by the geography or cost. Developing a predictive model for MASLD applicable to the general population with easily available predictors could facilitate early detection and timely intervention in the general population.

In this study, we aim to develop feasible and simplified MASLD prediction models using data from the 2017–2020 cycles of the National Health and Nutrition Examination Survey (NHANES) dataset, representative sample of the US general population. We compare the performance of various ML models and evaluate their ability to predict MASLD. Our findings enable to provide valuable insights into the clinical utility of a MASLD prediction framework among general populations, particularly in resource-limited primary healthcare settings.

## 2 | Methods

### 2.1 | Study Design and Populations

This cross-sectional study used data obtained from the 2017–2020 NHANES survey, which includes a wide range of clinical, demographic, and biochemical data [4, 12, 27]. All participants of the NHANES survey have provided written informed consent. The analysis protocol for this study was approved by the Ethics Committee of the First Affiliated Hospital of the Guangxi Medical University (IRB NO.: 2023-K61-01). Participants were not involved in the design, conduct, reporting, or dissemination plans of this research. The inclusion criteria were with complete liver elastography exam (i.e., fasting time of at least 3 h, 10 or more complete stiffness [E] measures, a liver stiffness interquartile [IQRe] range/median E < 30%, and available CAP measures). The exclusion criterium was aged < 20 years or excessive alcohol use (more than four drinks per day) [4, 12].

### 2.2 | Data Collection and Measure

Candidate variables included 32 predictors including demographic variables (age, gender, and education level), physical assessment (body weight, standing height, waist circumference, and hip circumference), disease history (hypertension, diabetes, and kidney diseases) or lifestyle behaviors (smoking and alcohol consumption), and laboratory values (ALT, AST, alkaline phosphatase [ALP], albumin, blood urea nitrogen [BUN], creatinine, glucose, GGT, lactate dehydrogenase [LDH], total bilirubin, white blood cell count [WBC], hemoglobin, lymphocyte number, segmented neutrophils, platelet count [PLT], direct high-density lipoprotein [HDL]-cholesterol, low-density lipoprotein [LDL]-cholesterol, triglyceride, creatine phosphokinase [CPK], and total cholesterol) from 2017–2020 NHANES. The BMI is not included due to the collinearity with weight and height which are considered the predictors.

### 2.3 | Diagnosis of MASLD

The primary outcome in this study was MASLD, defined as (1) CAP ≥ 280 dB/m among the participants with complete CAP exams by FibroScan (Echosens, France) in the NHANES transient elastography dataset; (2) present with at least one of five cardiometabolic risk factors: (a) BMI ≥ 25 kg/m$^2$, (b) diagnosed with Type 2 diabetes (DIQ010—Doctor told you have diabetes from the questionnaire data of NHANES), (c) high blood pressure (BPQ020—Ever told you had high blood pressure from the questionnaire data of NHANES), (d) plasma triglycerides ≥ 1.70 mmol/L, or (e) plasma HDL-cholesterol ≤ 1.0 mmol/L; and (3) without excessive alcohol use, defined as having > 4

alcoholic drinks (One drink means a 12-oz. beer, a 5-oz. glass of wine, or one and a half ounces of liquor.) [2, 28, 29].

### 2.4 | Data Preprocessing and Feature Selection

For the continuous variables, the missing values were imputed with median values, and all variables were standardized to be included in models. Variables, including age (20–29, 30–49, 50–69, 70 years), gender (female or male), race (Hispanic, non-Hispanic White, non-Hispanic Black, or non-Hispanic Asian/Other), education level (less than high school, high school graduate, or college or above), marital status (married/living with partner, widowed/divorced/separated, or never married/other), hypertension (yes or no), diabetes (yes or no), kidney diseases (yes or no), smoking (yes or no), and alcohol consumption (yes or no), were used as categorical variables. The missing value of these categorical variables was treated as a separate subgroup or combined with another subgroup.

For feature selection, we employed the least absolute shrinkage and selection operator (LASSO) regression and the Boruta algorithm. LASSO regression, which is effective in handling multicollinearity and enhancing model interpretability with L1 regularization, was applied to refine the variables by reducing the coefficients of less important predictors to zero. We utilized the glmnet package in R for LASSO regression of binary outcome data with 10-fold cross-validation. The Boruta algorithm, a robust and stable method built around RFs, was used to identify relevant variables by comparing the importance of each predictive variable. We used Boruta package in R to conduct this algorithm.

The overlapped variables by LASSO and Boruta algorithm were selected. Additionally, we set the number of predictable features into 10 to meet the demands of simplicity and feasibility at primary healthcare settings or self-evaluation by general populations. If the selected variables were closely related to each other (such as diagnosis of diabetes and glucose), only the more easily available one was included based on medical expertise and expert suggestions, and the feature selection process was redone to ensure eight variables at most to be selected in the final model.

### 2.5 | Development and Evaluation of ML Models

We applied the following ML models to predict MASLD with features selected by LASSO and Boruta algorithms: LR, XGBoost, bootstrap aggregating (Bagging), RF, naive Bayes (NB), light gradient boosting machine (LightGBM), decision tree (DT), and support vector machine (SVM) [30]. Each model was trained on 70% of the data, validated internally on another 20% of the data, and the remaining 10% was used for internal testing [20]. LR is a basic method for binary classification that estimates the probability of a given categorical outcome based on a set of predictor variables. In the LR model, glm function in Base R was used to get the results with classical LRs.

XGBoost is a gradient boosting method known for its high performance in large datasets, building multiple DTs to improve prediction accuracy. We set the learning rate of 10% and the maximum depth of a tree to four to limit the model's complexity and prevent overfitting.

To enhance the model's stability and accuracy, bagging was implemented which generates multiple resampled training subsets and combines predictions from individual base models. In this study, we employed a bagging model based on DTs using the bagging () function of the “ipred” package in R. The model was trained with the parameters for the number of bootstrap replications of 25 and coob value of “true” to enable out-of-bag estimates, providing an unbiased estimation of model performance.

RF, an ensemble approach that utilizes DTs as base classifiers, was also applied. RF constructs multiple trees by recursively partitioning the data, and final predictions are determined by a voting mechanism. The “randomForest” package in R was used with the Gini Index as the split criterion, 500 trees, and no restriction on maximum tree depth.

NB, a probabilistic classifier derived from Bayes' theorem under the assumption of feature independence, was implemented using the naiveBayes () function from the “e1071” package in R. This model is known for its robust performance with categorical and continuous data.

We further employed LightGBM, a gradient boosting DT framework optimized for efficiency and accuracy. The model was constructed using the lgb.train() function from “lightgbm” package in R to develop a binary classification model for predicting the risk of MASLD with the area under the curve (AUC) as the primary evaluation metric. The parameter of learning rate was set to 0.05, the complexity of individual trees to 31, and a total of boosting rounds to 100.

DT is a nonlinear model that splits the dataset into distinct groups based on specific threshold values of predictor variables. We employed a DT classifier using the rpart() function from the “rpart” package in R. The model was constructed using the classification method with the minimum split size set to 20 to get better generalization.

SVM, a supervised learning algorithm that identifies the optimal hyperplane to maximize the margin between class separation, was applied using the svmRadial method from the “caret” package in R with a Radial Basis Function kernel and five-fold cross-validation. This approach is particularly effective in high-dimensional feature spaces and when class boundaries are nonlinear [20].

The performance of all ML models was evaluated using ROC curves, AUC with 95% CI, sensitivity, specificity, accuracy, precision, and F1-score [20].

### 2.6 | Model Interpretability With SHAP

To interpret model outputs, we used Shapley additive explanations (SHAP), which quantify the contribution of each feature to the prediction [30]. SHAP calculates the average marginal contribution of individual features on the model's output by

comparing all feasible feature combinations, allowing for both global and local interpretation of model behavior [31]. For each feature, SHAP values represent its mean impact on model predictions across all observations.

### 2.7 | Statistical Analyses

Continuous variables were expressed as medians with interquartile ranges (IQRs) for non-normally distributed data. Categorical variables were presented as counts and percentages. Comparisons between participants with and without MASLD were performed using the chi-squared tests for categorical variables and Mann–Whitney *U* tests for continuous variables. A two-sided $p$ value $<0.05$ was considered statistically significant. All analyses were conducted using SAS Version 9.4 (SAS Institute Inc., Cary, North Carolina, USA) and R Version 4.4.3 (R Foundation for Statistical Computing, Vienna, Austria).

## 3 | Results

### 3.1 | Baseline Characteristics

A total of 6814 participants of 2017–2020 NHANES were included in this study. The flowchart of the study was shown in Figure 1. The median age (IQR) was 53 (37~65) years, female accounting for 53.0%. Among the participants, 58.9% were with college level or above, and 58.7% were married or living with partners.

MASLD was detected among 2611 (38.3%) participants. Between MASLD and non-MASLD participants, there were significant differences in age, gender, race, education level, marital status, history of hypertension, diabetes, and smoking (Table 1). The differences in height, weight, BMI, waist circumference, hip circumference, ALT, AST, ALP, albumin, BUN, glucose, GGT, LDH, total bilirubin, WBC, hemoglobin, PLT, triglyceride, CPK, and total cholesterol between MASLD and non-MASLD groups were also statistically significant. Baseline characteristics of patients in the training, internal validation, and internal testing cohorts were shown in Table S1. In Table S1, the baseline characteristics, including frequency distributions of age, gender, race, education level, marital status, hypertension, diabetes, kidney disease, smoking, drinking, and MASLD, were presented by the groups of training, internal validation, and internal testing cohorts. The median and IQR of anthropometric and biochemical variables were also shown.

### 3.2 | Feature Selection

The predictable variables selected by LASSO and Boruta algorithms, after adjustment with clinical expert suggestions, were body weight, standing height, waist circumference, ALT, AST, GGT, and diagnosis of diabetes (Have any doctor told you have diabetes?). (Figure S1 showed the features selected by LASSO and Boruta algorithms. In Figure S1, both methods identified weight, height, waist circumference, ALT, AST, GGT, and diagnosis of diabetes [DB] as predictors. In addition, LASSO algorithm exclusively selected PLT and BUN, while Boruta algorithm uniquely identified hip circumference.)

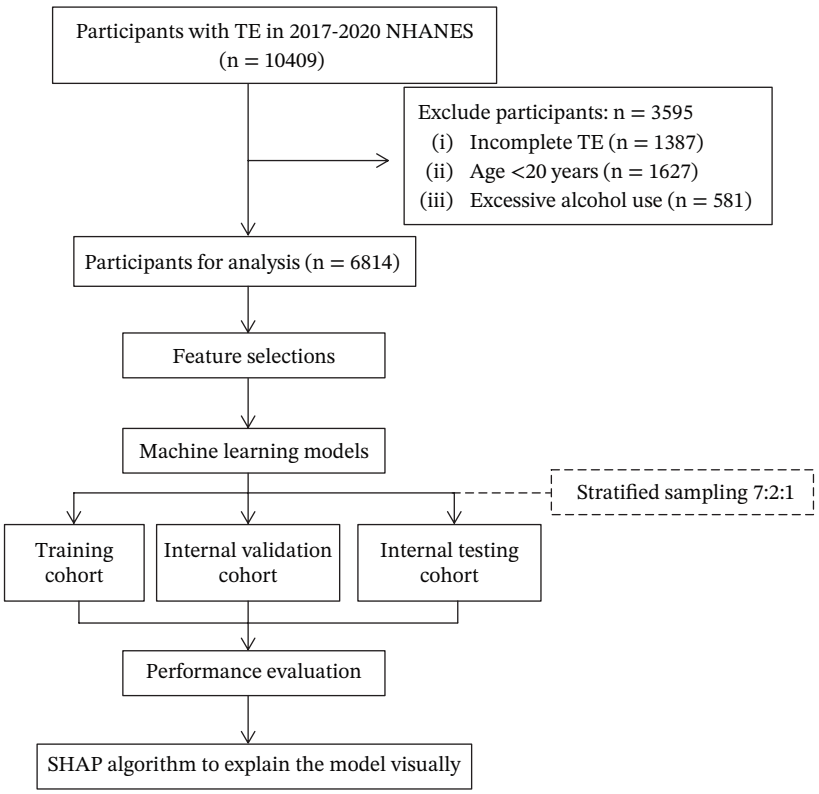



**FIGURE 1** | Flowchart of the STUDY. NHANES: National Health and Nutrition Examination Survey; SHAP: SHapley additive explanations; TE: transient elastography.

**TABLE 1** | Comparison of general characteristics of NHANES 2017–2020 cohorts.

| Characteristics | Total $N = 6814$ | Non-MASLD $n = 4203$ | MASLD $n = 2611$ | $p$ value |
|---|---|---|---|---|
| Age (years) | 53 (37, 65) | 50 (34, 64) | 56 (42, 66) | < 0.001 |
| 20–29 | 950 (13.9%) | 735 (17.5%) | 215 (8.2%) | |
| 30–49 | 2107 (30.9%) | 1342 (31.9%) | 765 (29.3%) | |
| 50–69 | 2584 (37.9%) | 1410 (33.5%) | 1174 (45.0%) | |
| 70– | 1173 (17.2%) | 716 (17.0%) | 457 (17.5%) | |
| Gender | | | | < 0.001 |
| Male | 3200 (47.0%) | 1827 (43.5%) | 1373 (52.6%) | |
| Female | 3614 (53.0%) | 2376 (56.5%) | 1238 (47.4%) | |
| Race | | | | < 0.001 |
| Hispanic | 1423 (20.9%) | 769 (18.3%) | 654 (25.0%) | |
| Non-Hispanic White | 1219 (17.9%) | 791 (18.8%) | 428 (16.4%) | |
| Non-Hispanic Black | 1862 (27.3%) | 1273 (30.3%) | 589 (22.6%) | |
| Non-Hispanic Asian/Other | 2310 (33.9%) | 1370 (32.6%) | 940 (36.0%) | |
| Education level | | | | 0.013 |
| Less than high school | 1598 (23.5%) | 961 (22.9%) | 637 (24.4%) | |
| High school graduate | 1202 (17.6%) | 709 (16.9%) | 493 (18.9%) | |
| College or above | 4014 (58.9%) | 2533 (60.3%) | 1481 (56.7%) | |
| Marital status | | | | < 0.001 |
| Married/living with partner | 4001 (58.7%) | 2351 (55.9%) | 1650 (63.2%) | |
| Widowed/divorced/separated | 1286 (18.9%) | 904 (21.5%) | 382 (14.6%) | |
| Never married/other | 1527 (22.4%) | 948 (22.6%) | 579 (22.2%) | |
| Hypertension history | | | | < 0.001 |
| Yes | 2625 (38.5%) | 1329 (31.6%) | 1296 (49.6%) | |
| No | 4189 (61.5%) | 2874 (68.4%) | 1315 (50.4%) | |
| Diabetes history | | | | < 0.001 |
| Yes | 1019 (15.0%) | 384 (9.1%) | 635 (24.3%) | |
| No | 5795 (85.0%) | 3819 (90.9%) | 1976 (75.7%) | |
| Kidney disease history | | | | 0.282 |
| Yes | 259 (3.8%) | 151 (3.6%) | 108 (4.1%) | |
| No | 4120 (60.5%) | 2592 (61.7%) | 1528 (58.5%) | |
| Smoking | | | | 0.011 |
| Yes | 2694 (39.5%) | 1611 (38.3%) | 1083 (41.5%) | |
| No | 4120 (60.5%) | 2592 (61.7%) | 1528 (58.5%) | |
| Drinking > once a week | | | | 0.003 |
| Yes | 1595 (23.4%) | 1034 (24.6%) | 561 (21.5%) | |
| No | 5219 (76.6%) | 3169 (75.4%) | 2050 (78.5%) | |
| Height (cm) | 166.0 (159.0, 173.5) | 165.6 (158.7, 173.1) | 166.4 (159.6, 174.2) | < 0.001 |
| Weight (kg) | 79.2 (67.1, 93.9) | 72.7 (62.1, 84.4) | 90.8 (77.9, 106.3) | < 0.001 |

(Continues)

**TABLE 1 | (CONTINUED)**

| Characteristics | Total $N = 6814$ | Non-MASLD $n = 4203$ | MASLD $n = 2611$ | $p$ value |
|---|---|---|---|---|
| Body mass index (kg/m$^2$) | 28.6 (24.7, 33.2) | 26.2 (23.2, 30.0) | 32.2 (28.6, 37.2) | < 0.001 |
| Waist circumference (cm) | 98.9 (88.9, 109.7) | 93.2 (83.9, 101.8) | 108.3 (98.9, 119.2) | < 0.001 |
| Hip circumference (cm) | 104.4 (97.2, 113.5) | 101.1 (94.6, 108.3) | 110.1 (103.4, 121.3) | < 0.001 |
| ALT (U/L) | 18 (13, 24) | 16 (12, 21) | 20 (16, 30) | < 0.001 |
| AST (U/L) | 19 (16, 23) | 19 (16, 22) | 19 (16, 24) | < 0.001 |
| ALP (IU/L) | 74 (62, 88) | 73 (60, 85) | 76 (66, 92) | < 0.001 |
| Albumin (g/L) | 41 (39, 43) | 41 (39, 43) | 41 (38, 42) | < 0.001 |
| Blood urea nitrogen (mmol/L) | 5.00 (3.93, 6.07) | 5.00 (3.93, 6.07) | 5.00 (4.28, 6.07) | < 0.001 |
| Creatinine ($\mu$mol/L) | 74.26 (62.76, 87.52) | 74.26 (62.76, 87.52) | 74.26 (62.76, 88.40) | 0.318 |
| Glucose (mmol/L) | 5.16 (4.83, 5.66) | 5.11 (4.77, 5.44) | 5.38 (5.05, 6.22) | < 0.001 |
| GGT (IU/L) | 21 (15, 30) | 19 (13, 26) | 24 (19, 38) | < 0.001 |
| LDH (IU/L) | 154 (138, 173) | 154 (137, 171) | 154 (141, 174) | < 0.001 |
| Total bilirubin ($\mu$mol/L) | 6.84 (5.13, 8.55) | 6.84 (5.13, 8.55) | 6.84 (5.13, 8.55) | 0.035 |
| WBC (1000 cells/$\mu$L) | 6.8 (5.6, 8.2) | 6.7 (5.4, 7.9) | 7.1 (6.1, 8.7) | < 0.001 |
| Hemoglobin (g/dL) | 14.1 (13.1, 14.9) | 13.9 (13.0, 14.8) | 14.2 (13.3, 15.2) | < 0.001 |
| Lymphocyte number (1000 cells/$\mu$L) | 2.1 (1.7, 2.6) | 2.1 (1.6, 2.5) | 2.2 (1.8, 2.7) | < 0.001 |
| Segmented neutrophils (1000 cell/$\mu$L) | 3.9 (3.0, 4.9) | 3.8 (2.9, 4.7) | 4.1 (3.3, 5.2) | < 0.001 |
| Platelet count (1000 cells/$\mu$L) | 239 (204, 279) | 239 (202, 277) | 239 (207, 284) | < 0.001 |
| Direct HDL-cholesterol (mmol/L) | 1.32 (1.11, 1.58) | 1.40 (1.22, 1.68) | 1.22 (1.03, 1.40) | < 0.001 |
| Triglyceride (mmol/L) | 2.741 (2.741, 2.741) | 2.741 (2.741, 2.741) | 2.741 (2.741, 2.741) | 0.041 |
| CPK (IU/L) | 1.010 (1.010, 1.010) | 1.010 (0.881, 1.010) | 1.010 (1.010, 1.253) | < 0.001 |
| Total cholesterol (mmol/L) | 4.73 (4.14, 5.40) | 4.73 (4.11, 5.33) | 4.73 (4.14, 5.46) | 0.002 |

*Note:* The categorical data were presented with frequencies and percentages. Continuous variables were presented with median (interquartile range). $p$ values were generated with chi-squared tests and nonparametric tests for categorical and continuous variable, respectively.

Abbreviations: ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; CPK, creatine phosphokinase; GGT, gamma glutamyl transferase; HDL, high-density lipoprotein; LDH, lactate dehydrogenase; MASLD, metabolic dysfunction-associated steatotic liver disease; WBC, white blood cell count.

### 3.3 | Assessment of the Accuracy of ML Models for the Diagnosis of MASLD

The performances of all the ML models were evaluated with AUC, accuracy, sensitivity, specificity, precision, and F1-score by internal validation cohort and internal testing cohort (Table 2). The LR, XGBoost, RF, and LightGBM models achieved higher AUC (0.841, 0.837, 0.838, and 0.842) than other models in the internal validation cohort, indicating strong predictive ability for MASLD with the selected features. In the internal testing cohort, the LR, XGBoost, RF, and LightGBM models achieved higher AUC (0.817, 0.815, 0.813, and 0.825) than other models. The Bagging, NB, DT, and SVM models also achieved AUCs > 0.75 (Figure 2).

### 3.4 | Visualization by SHAP

Figure 3a presents the feature importance ranking for the XGBoost model derived from SHAP values. Predictors were ordered by the mean absolute SHAP value, reflecting their relative contribution to the model's output. The SHAP summary plot (Figure 3b) further depicts the direction and magnitude of each feature's effect on model predictions. Higher SHAP values for specific predictors indicated an elevated risk of MASLD. Waist circumference, ALT, weight, height, and GGT emerged as the five most influential variables. For instance, patients exhibiting aberrant waist circumference were found to be predisposed to a higher incidence of MASLD, compared with those with normal waist circumference. Noninvasive liver-related biochemical markers, such as ALT, AST, and GGT, also proved to be significant predictors of MASLD.

## 4 | Discussion

This study demonstrated the potential of various ML models in predicting MASLD using commonly available anthropometric and biochemical variables via NHANES 2017–2020. The selected seven common features, including body weight, height, waist circumference, diabetes history, ALT, AST, and GGT,

**TABLE 2** | Performance of machine learning models for predicting MASLD in NHANES 2017–2020 cohorts.

| Cohort | Model | AUC | 95%CI AUC | Optimal cutoff | Accuracy | Sensitivity | Specificity | Precision | F1-score |
|---|---|---|---|---|---|---|---|---|---|
| Internal validation cohort | LR | 0.841 | 0.820~0.861 | 0.373 | 0.767 | 0.761 | 0.771 | 0.674 | 0.715 |
| | XGBoost | 0.837 | 0.816~0.858 | 0.389 | 0.761 | 0.785 | 0.745 | 0.657 | 0.716 |
| | Bagging | 0.815 | 0.792~0.837 | 0.420 | 0.742 | 0.728 | 0.751 | 0.645 | 0.684 |
| | RF | 0.838 | 0.817~0.859 | 0.327 | 0.750 | 0.866 | 0.679 | 0.626 | 0.727 |
| | NB | 0.814 | 0.792~0.837 | 0.181 | 0.747 | 0.751 | 0.745 | 0.647 | 0.695 |
| | LightGBM | 0.842 | 0.822~0.863 | 0.392 | 0.767 | 0.785 | 0.756 | 0.667 | 0.721 |
| | DT | 0.796 | 0.773~0.820 | 0.355 | 0.740 | 0.782 | 0.714 | 0.630 | 0.697 |
| | SVM | 0.828 | 0.805~0.850 | 0.380 | 0.773 | 0.716 | 0.808 | 0.699 | 0.708 |
| Internal testing cohort | LR | 0.817 | 0.786~0.848 | 0.366 | 0.743 | 0.709 | 0.764 | 0.651 | 0.679 |
| | XGBoost | 0.815 | 0.784~0.847 | 0.430 | 0.756 | 0.732 | 0.771 | 0.666 | 0.697 |
| | Bagging | 0.797 | 0.764~0.830 | 0.300 | 0.705 | 0.835 | 0.624 | 0.580 | 0.684 |
| | RF | 0.813 | 0.781~0.844 | 0.305 | 0.711 | 0.839 | 0.631 | 0.586 | 0.690 |
| | NB | 0.786 | 0.752~0.820 | 0.192 | 0.724 | 0.686 | 0.748 | 0.628 | 0.656 |
| | LightGBM | 0.825 | 0.794~0.855 | 0.313 | 0.722 | 0.858 | 0.638 | 0.596 | 0.703 |
| | DT | 0.768 | 0.732~0.804 | 0.355 | 0.714 | 0.747 | 0.693 | 0.602 | 0.667 |
| | SVM | 0.799 | 0.764~0.833 | 0.346 | 0.756 | 0.701 | 0.790 | 0.675 | 0.688 |

Abbreviations: AUC, area under the curve; Bagging, bootstrap aggregating; DT, decision tree; LightGBM, light gradient boosting machine; LR, logistic regression; MASLD, metabolic dysfunction-associated steatotic liver disease; NB, naive Bayes; NHANES, National Health and Nutrition Examination Survey; RF, random forest; SVM, support vector machines; XGBoost, extreme gradient boosting.

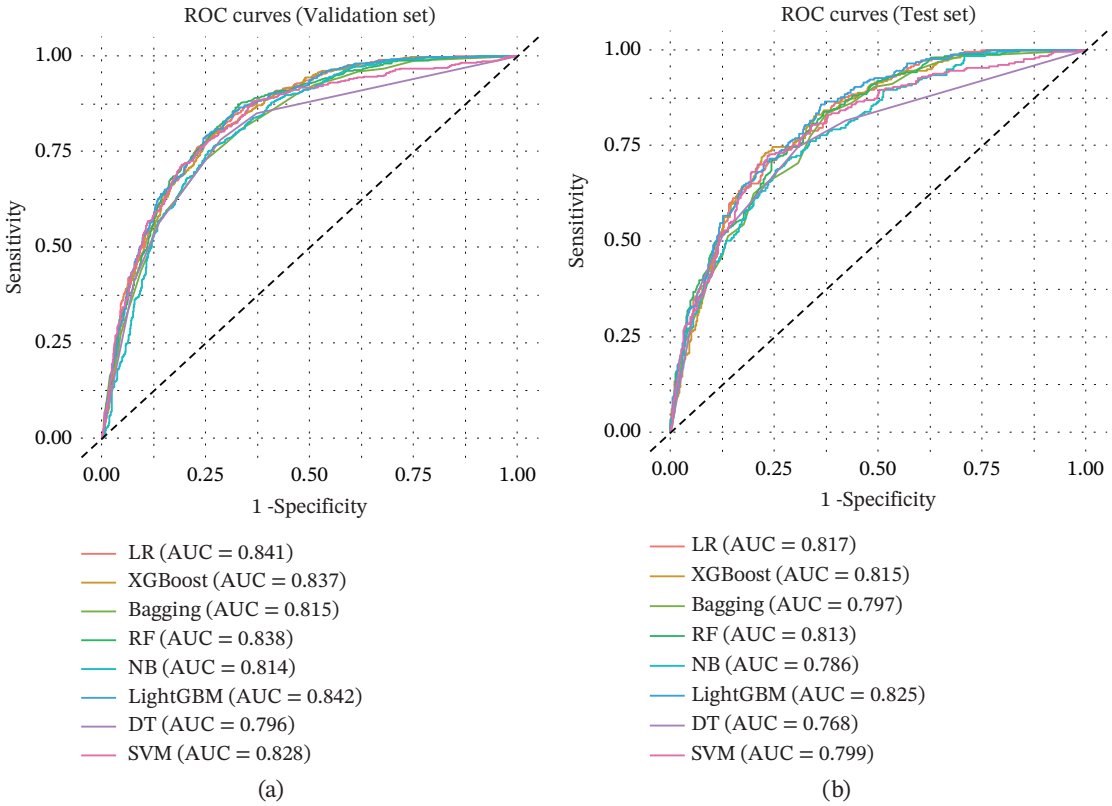



**FIGURE 2** | Prediction performance by AUCs of (a) internal validation and (b) internal testing cohorts for the machine learning models for predicting MASLD in NHANES 2017–2020 cohorts. Abbreviations: AUC, area under the curve; Bagging, bootstrap aggregating; DT, decision tree; LightGBM, light gradient boosting machine; LR, logistic regression; MASLD, metabolic dysfunction-associated steatotic liver disease; NB, naive Bayes; NHANES, National Health and Nutrition Examination Survey; RF, random forest; ROC, receiver operating characteristic; SVM, support vector machines; XGBoost, extreme gradient boosting.

showed significant accuracy of the ML prediction models and high potential in wide use at resource-limited primary healthcare setting or self-evaluation scenarios.

The predictive power of ML methods highlights the benefits of incorporating anthropometric, clinical, and biochemical variables. The ML models by similar studies [32, 33] showed better efficacy than the traditional noninvasive indices such as FIB-4, APRI, and Forns. Noninvasive biomarkers, such as weight, waist circumference, and diabetes, proved to be significant predictors of MASLD. These findings align with previous research that has highlighted the role of metabolic factors in MASLD development [2, 12, 19, 34]. The inclusion of liver-related biomarkers, such as ALT, AST, and GGT, further enhances the ability to predict MASLD, as these biochemical variables are commonly available and reflect liver dysfunction and systemic inflammation [33]. In this study, similarly in other published results, the waist circumference is crucial to predict MASLD [32, 35].

Additionally, this study did not use other advanced biochemical markers, such as Apolipoprotein A-1, hyaluronic acid, or tissue inhibitor of metalloproteinase 1, which may hinder the wide use of the prediction model in primary healthcare setting [10]. So, the ML models from this study possess high potential for wide use in resource-limited settings. In resource-limited settings where VCTE or TE is impractical, the ML model developed in our study may function as a low-cost screening tool for MASLD. Only high-risk individuals identified by the algorithm are referred for confirmatory VCTE or MRI-PDFF tests, thus optimizing the use of limited imaging capacity and cutting down on patient expenses. The diagnosis of diabetes can also be represented by glucose-related indicators such as fasting blood glucose or impaired glucose tolerance. However, self-reported diabetes history or diagnosis from medical records is more convenient and widely available. Therefore, in the broader application of ML models, this predictor can be flexibly handled [2].

To our knowledge, this study is among the early studies to utilize ML models with feasible and simple anthropometric and biochemical predictors to predict the risk of MASLD in the general population of the United States. The models of LR, XGBoost, and LightGBM with simple predictors of body weight, standing height, waist circumference, ALT, AST, GGT, and diagnosis with diabetes outperform other models. There were some limitations for the study. First, the diagnosis of MASLD in our study was mainly based on FibroScan measurements and excluded the participants with excessive alcohol use, which may introduce certain errors due to factors like optimal cutoff values, operator variability, and patient conditions, and does not reflect the full metabolic features of MASLD, such as glucose and lipid abnormalities. Moreover, we did not use different criteria for HDL-cholesterol or alcohol use. Second, the models have not been validated using external

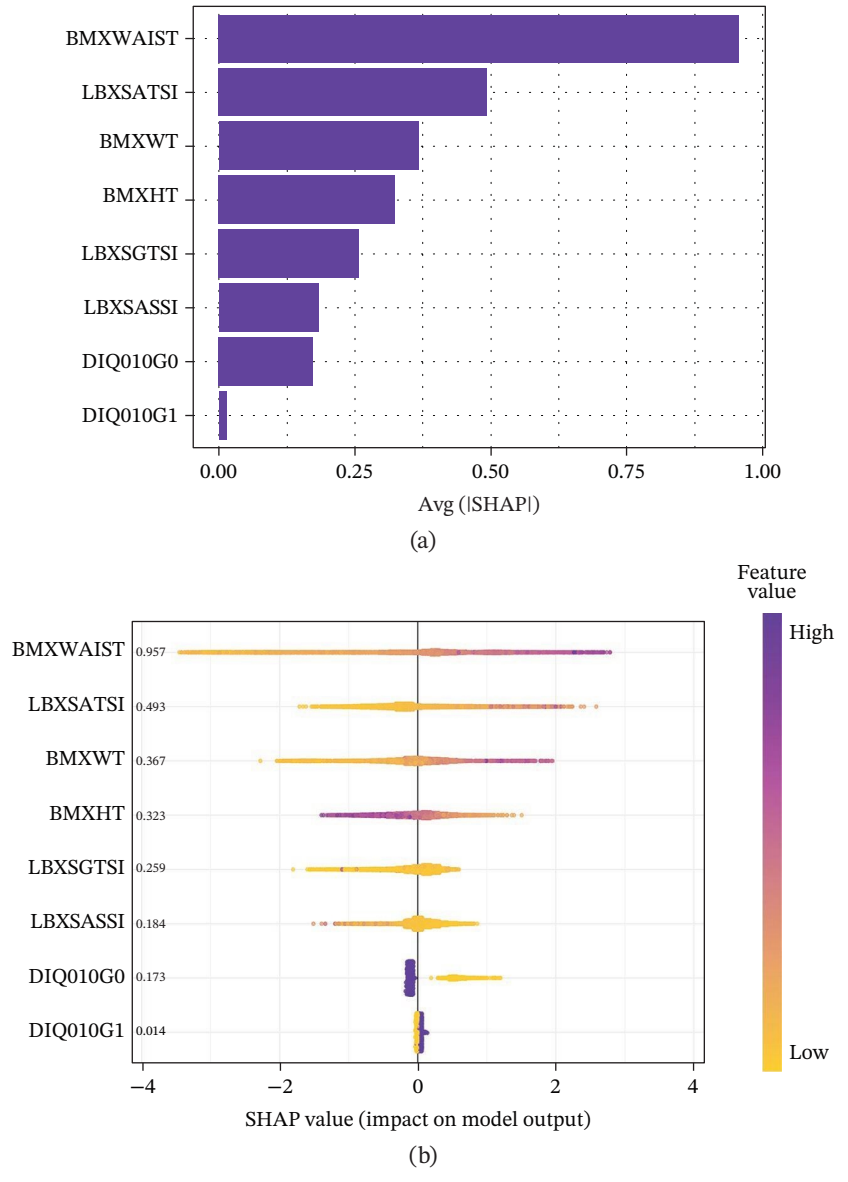



**FIGURE 3** | (a) SHAP summary plot showing the distribution of the SHAP values of each feature; (b) SHAP feature importance shown according to the mean absolute SHAP value of each feature in NHANES 2017–2020 cohorts.

datasets. Therefore, its generalizability to other regions and populations may be confined. Future studies should incorporate external various databases, such as the NHANES 2021–2023 dataset, with some biochemical variables not available and requiring imputation, real-world data from hospitals, or physical examination centers. Third, the study's cross-sectional design limits the ability to infer causal relationships or predict future risk of MASLD. However, the samples size and data quality from NHANES can guarantee the robustness and stability of the results [27].

## 5 | Conclusion

Our study demonstrated that the ML models with simple anthropometric and biochemical measurements effectively identify MASLD in general populations. These findings provide valuable insights for clinical applications, as they can aid in early detection, less invasive screening, and personalized treatment and intervention planning for patients at risk of MASLD in a large general population. Future research should focus on further validating these models with external datasets, incorporating additional common biomarkers.

### Author Contributions

Y.X. and H.Y. conceived of and designed the study. S.D., J.D., and H.Y. performed the data curation and analysis and contributed to interpreting the results. Y.X. verified the analytical methods and results. S.D. drafted the manuscript. Y.X., H.Y., and J.D. revised the manuscript. Y.X. acts as a guarantor responsible for the overall content. S.D. and H.Y. have contributed equally.

### Acknowledgments

We gratefully acknowledge the participants of the NHANES survey and the dedicated staff for their roles in study design, subject recruitment, data collection, and technical assistance.

## Funding

This work was supported by the Youth Natural Science Foundation of Guangxi (2024GXNSFBA010067 to Y.X.), the Clinical Research "Climbing" Program of the First Affiliated Hospital of Guangxi Medical University (YYZS2023015 to Y.X.), and the National Natural Science Foundation of China (82404382 to Y.X.).

## Disclosure

All authors have discussed the results, read, and approved the final manuscript. The funders had no role in study design, data collection, analysis, interpretation, or writing of the manuscript.

## Ethics Statement

The analysis protocol for this study was approved by the Ethics Committee of the First Affiliated Hospital of Guangxi Medical University (IRB NO.: 2023-K61-01).

## Conflicts of Interest

The authors declare no conflicts of interest.

## Data Availability Statement

The NHANES data are publicly available at https://wwwn.cdc.gov/nchs/nhanes/search/datapage.aspx?Component=Examination%26Cycle=2017-2020.

## References

1. J. Li, B. Zou, Y. H. Yeo, et al., "Prevalence, Incidence, and Outcome of Non-Alcoholic Fatty Liver Disease in Asia, 1999-2019: A Systematic Review and Meta-Analysis," *Lancet Gastroenterology & Hepatology* 4 (2019): 389–398.

2. M. Kalligeros, S. Vassilopoulos, A. Vassilopoulos, et al., "Prevalence and Risk Factors of Nonalcoholic Fatty Liver Disease, High-Risk Nonalcoholic Steatohepatitis, and Fibrosis Among Lean United States Adults: NHANES 2017-2020," *Annals of Gastroenterology* 36 (2023): 670–677, https://doi.org/10.20524/aog.2023.0840.

3. M. Israelsen, S. Francque, E. A. Tsochatzis, and A. Krag, "Steatotic Liver Disease," *Lancet* 404 (2024): 1761–1778.

4. M. Noureddin, F. Ntanios, D. Malhotra, et al., "Predicting NAFLD Prevalence in the United States Using National Health and Nutrition Examination Survey 2017-2018 Transient Elastography Data and Application of Machine Learning," *Hepatology Communications* 6, no. 7 (2022): 1537–1548, https://doi.org/10.1002/hep4.1935.

5. K. S. Kim, S. Hong, K. Han, and C. Y. Park, "Association of Non-Alcoholic Fatty Liver Disease With Cardiovascular Disease and All Cause Death in Patients With Type 2 Diabetes Mellitus: Nationwide Population Based Study," *BMJ* 384 (2024): e076388, https://doi.org/10.1136/bmj-2023-076388.

6. R. Song, Z. Li, Y. Zhang, J. Tan, and Z. Chen, "Comparison of NAFLD, MAFLD and MASLD Characteristics and Mortality Outcomes in United States Adults," *Liver International* 44, no. 4 (2024): 1051–1060, https://doi.org/10.1111/liv.15856.

7. J. M. Paik, P. Golabi, Y. Younossi, A. Mishra, and Z. M. Younossi, "Changes in the Global Burden of Chronic Liver Diseases From 2012 to 2017: The Growing Impact of NAFLD," *Hepatology* 72, no. 5 (2020): 1605–1616.

8. D. Wang, J. Miao, L. Zhang, and L. Zhang, "Research Advances in the Diagnosis and Treatment of MASLD/MASH," *Annals of Medicine* 57, no. 1 (2025): 2445780, https://doi.org/10.1080/07853890.2024.2445780.

9. E. Vilar-Gomez, and N. Chalasani, "Non-Invasive Assessment of Non-Alcoholic Fatty Liver Disease: Clinical Prediction Rules and Blood-Based Biomarkers," *Journal of Hepatology* 68, no. 2 (2018): 305–315, https://doi.org/10.1016/j.jhep.2017.11.013.

10. F. Abdelhameed, C. Kite, L. Lagojda, et al., "Non-Invasive Scores and Serum Biomarkers for Fatty Liver in the Era of Metabolic Dysfunction-associated Steatotic Liver Disease (MASLD): A Comprehensive Review From NAFLD to MAFLD and MASLD," *Current Obesity Reports* 13 (2024): 510–531, https://doi.org/10.1007/s13679-024-00574-z.

11. Y. Xuan, D. Wu, Q. Zhang, Z. Yu, J. Yu, and D. Zhou, "Elevated ALT/AST Ratio as a Marker for NAFLD Risk and Severity: Insights From a Cross-Sectional Analysis in the United States," *Frontiers in Endocrinology* 15 (2024): 1457598, https://doi.org/10.3389/fendo.2024.1457598.

12. A. Unalp-Arida, and C. E. Ruhl, "Transient Elastography-Assessed Hepatic Steatosis and Fibrosis Are Associated With Body Composition in the United States," *Clinical Gastroenterology and Hepatology* 20 (2022): e808–e830, https://doi.org/10.1016/j.cgh.2021.02.009.

13. Y. Xie, K. Huang, X. Zhang, et al., "Association of Serum Uric Acid-to-High-Density Lipoprotein Cholesterol Ratio With Non-Alcoholic Fatty Liver Disease in American Adults: A Population-Based Analysis," *Frontiers in Medicine* 10 (2023): 1164096, https://doi.org/10.3389/fmed.2023.1164096.

14. M. Chang, Z. Shao, and G. Shen, "Association Between Triglyceride Glucose-Related Markers and the Risk of Metabolic-Associated Fatty Liver Disease: A Cross-Sectional Study in Healthy Chinese Participants," *BMJ Open* 13, no. 5 (2023): e070189.

15. H. Chen, J. Zhang, X. Chen, et al., "Development and Validation of Machine Learning Models for MASLD: Based on Multiple Potential Screening Indicators," *Frontiers in Endocrinology* 15 (2025): 1449064, https://doi.org/10.3389/fendo.2024.1449064.

16. Q. Chen, P. Hu, X. Hou, et al., "Association Between Triglyceride-Glucose Related Indices and Mortality Among Individuals With Non-Alcoholic Fatty Liver Disease or Metabolic Dysfunction-Associated Steatotic Liver Disease," *Cardiovascular Diabetology* 23 (2024): 232, https://doi.org/10.1186/s12933-024-02343-7.

17. J. Lee, M. Westphal, Y. Vali, et al., "Machine Learning Algorithm Improves the Detection of NASH (NAS-Based) and At-Risk NASH: A Development and Validation Study," *Hepatology* 78, no. 1 (2023): 258–271, https://doi.org/10.1097/HEP.0000000000000364.

18. C. H. Lu, W. Wang, Y. C. Li, et al., "Machine Learning Models for Predicting Significant Liver Fibrosis in Patients With Severe Obesity and Nonalcoholic Fatty Liver Disease," *Obesity Surgery* 34 (2024): 4393–4404, https://doi.org/10.1007/s11695-024-07548-z.

19. P. Sripongpun, W. R. Kim, A. Mannalithara, et al., "The Steatosis-Associated Fibrosis Estimator (SAFE) Score: A Tool to Detect Low-Risk NAFLD in Primary Care," *Hepatology* 77, no. 1 (2023): 256–267, https://doi.org/10.1002/hep.32545.

20. G. Zhu, Y. Song, Z. Lu, et al., "Machine Learning Models for Predicting Metabolic Dysfunction-Associated Steatotic Liver Disease Prevalence Using Basic Demographic and Clinical Characteristics," *Journal of Translational Medicine* 23 (2025): 381, https://doi.org/10.1186/s12967-025-06387-5.

21. K. Nabrdalik, H. Kwiendacz, K. Irlik, et al., "Machine Learning Identifies Metabolic Dysfunction-Associated Steatotic Liver Disease in Patients With Diabetes Mellitus," *Journal of Clinical Endocrinology \& Metabolism* 109 (2024): 2029–2038, https://doi.org/10.1210/clinem/dgae060.

22. S. Hassoun, C. Bruckmann, S. Ciardullo, G. Perseghin, F. Di Gaudio, and F. Broccolo, "Setting Up of a Machine Learning Algorithm for the Identification of Severe Liver Fibrosis Profile in the General US Population Cohort," *International Journal of Medical Informatics* 170 (2023): 104932, https://doi.org/10.1016/j.ijmedinf.2022.104932.

23. X. Chen, G. B. Goh, J. Huang, et al., "Validation of Non-Invasive Fibrosis Scores for Predicting Advanced Fibrosis in Metabolic-Associated

Fatty Liver Disease," *Journal of Clinical and Translational Hepatology* 10 (2022): 589–594, https://doi.org/10.14218/JCTH.2021.00311.

24. F. Razmpour, R. Daryabeygi-Khotbehsara, D. Soleimani, et al., "Application of Machine Learning in Predicting Non-Alcoholic Fatty Liver Disease Using Anthropometric and Body Composition Indices," *Scientific Reports* 13 (2023): 4942, https://doi.org/10.1038/s41598-023-32129-y.

25. B. Njei, E. Osta, N. Njei, Y. A. Al-Ajlouni, and J. K. Lim, "An Explainable Machine Learning Model for Prediction of High-Risk Nonalcoholic Steatohepatitis," *Scientific Reports* 14 (2024): 8589, https://doi.org/10.1038/s41598-024-59183-4.

26. W. Ji, M. Xue, Y. Zhang, H. Yao, and Y. Wang, "A Machine Learning Based Framework to Identify and Classify Non-alcoholic Fatty Liver Disease in a Large-Scale Population," *Frontiers in Public Health* 10 (2022): 846118, https://doi.org/10.3389/fpubh.2022.846118.

27. V. Charu, J. W. Liang, A. Mannalithara, A. Kwong, L. Tian, and W. R. Kim, "Benchmarking Clinical Risk Prediction Algorithms With Ensemble Machine Learning for the Noninvasive Diagnosis of Liver Fibrosis in NAFLD," *Hepatology* 80, no. 5 (2024): 1184–1195, https://doi.org/10.1097/HEP.0000000000000908.

28. European Association for the Study of the Liver, "EASL Clinical Practice Guidelines on Non-Invasive Tests for Evaluation of Liver Disease Severity and Prognosis - 2021 Update," *Journal of Hepatology* 75, no. 3 (2021): 659–689, https://doi.org/10.1016/j.jhep.2021.05.025.

29. M. E. Rinella, J. V. Lazarus, V. Ratziu, et al., "A Multisociety Delphi Consensus Statement on New Fatty Liver Disease Nomenclature," *Hepatology* 78 (2023): 1966–1986, https://doi.org/10.1097/HEP.0000000000000520.

30. T. Sun, J. Liu, H. Yuan, X. Li, and H. Yan, "Construction of a Risk Prediction Model for Lung Infection After Chemotherapy in Lung Cancer Patients Based on the Machine Learning Algorithm," *Frontiers in Oncology* 14 (2024): 1403392, https://doi.org/10.3389/fonc.2024.1403392.

31. O. O. Bifarin, "Interpretable Machine Learning With Tree-Based Shapley Additive Explanations: Application to Metabolomics Datasets for Binary Classification," *PLoS One* 18, no. 5 (2023): e0284315, https://doi.org/10.1371/journal.pone.0284315.

32. L. A. van Kleef, S. M. Francque, J. E. Prieto-Ortiz, et al., "Metabolic Dysfunction-Associated Fibrosis 5 (MAF-5) Score Predicts Liver Fibrosis Risk and Outcome in the General Population With Metabolic Dysfunction," *Gastroenterology* 167 (2024): 357–367.e359, https://doi.org/10.1053/j.gastro.2024.03.017.

33. C. F. Huang, P. C. Liang, C. W. Wang, et al., "Performance of Noninvasive Seromarkers in Predicting Liver Fibrosis Among MAFLD Patients With or Without Viral Hepatitis," *Kaohsiung Journal of Medical Sciences* 40 (2024): 374–383, https://doi.org/10.1002/kjm2.12804.

34. M. Serra-Burriel, A. Juanola, F. Serra-Burriel, et al., "Development, Validation, and Prognostic Evaluation of a Risk Score for Long-Term Liver-Related Outcomes in the General Population: A Multicohort Study," *Lancet* 402 (2023): 988–996.

35. Y. Shen, Y. Wu, M. Fu, K. Zhu, and J. Wang, "Association Between Weight-Adjusted-Waist Index With Hepatic Steatosis and Liver Fibrosis: A Nationally Representative Cross-Sectional Study From NHANES 2017 to 2020," *Frontiers in Endocrinology* 14 (2023): 1159055, https://doi.org/10.3389/fendo.2023.1159055.

**Supporting Information**

Additional supporting information can be found online in the Supporting Information section. *(Supporting Information)* Figure S1 Feature selection results using LASSO and Boruta algorithms. Table S1: Baseline characteristics and outcomes of patients in the training, internal validation, and internal testing cohorts in 2017–2020 NHANES.